Luke Brzozowski

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Result-driven with 3+ years work experience in Engineering, seeking positions in Software Engineering or Embedded Systems.

EDUCATION

The University of Michigan - Ann Arbor | *Bachelor of Science in Engineering in Computer Engineering* Dec 2025

Courses: Computer Vision, Computer Security, Computational Linear Algebra, Data Structures and Algorithms, Computer Architecture, Embedded System Design, Embedded Control, Signals and Systems, Digital Logic Design

EXPERIENCE

Agentic Fruit, *Technical Co-Founder* | Ann Arbor, MI May 2025 – Present

- Built multi-agent AI coding system (**TypeScript/LangGraph/Next.js**) with 90% code accuracy and 60% development time reduction
- Implemented RAG-based system using Claude and prompt engineering techniques, achieving 92% user query satisfaction
- Developed AI hotel administration platform with automated full-stack code generation (QloApps/Stripe/ElevenLabs integration)

Robert Bosch, *Systems Engineering Intern* | Plymouth, MI May 2024 – Jan 2025

- Led end-to-end development of voice-activated ADAS using **C++** and **Raspberry Pi**, achieving 90% intent classification accuracy through Google Dialogflow optimizations and 200ms response time, with reliable vehicle integration through CAN bus protocols. Delivered market-ready system showcased at CES 2025 to 10,000+ attendees, receiving positive feedback from automotive executives
- Introduced and developed automation solutions using **Python** (Selenium, pandas) and Excel scripting to streamline data collection, processing, and bulk database updates, resulting in 90% reduction in manual labor for associated tasks

UTAC Inc., *Software Development Intern* | Northville Township, MI May 2023 – Aug 2023

- Utilized LabVIEW to enhance software functionality by integrating email, data acquisition, and data export features
- Developed live power monitoring/logging system for test stands with cost reporting module for usage analytics

UoM Formula SAE Electric, *Autonomous Team Member* | Ann Arbor, MI Aug 2023 – Jan 2024

- Developed trajectory tracking algorithm for autonomous vehicle, increasing tracking accuracy by 25% over previous versions

UoM Formula SAE Combustion, *Powertrain Team Member* | Dearborn, MI Aug 2022 – Apr 2023

- Designed and simulated parts using SOLIDWORKS. Developed requirements, manufactured parts, and conducted validation testing
- Rebuilt engine and conducted dynamometer analysis for performance tuning

Goodlyfe LLC, *Manager* | Van Buren Twp, MI Jan 2021 – Jan 2022

- Streamlined production processes and implemented data-driven scheduling optimizations, saving $5000 per month

PROJECTS

Piazza Post Categorizer | *Machine Learning (ML), C++, Binary Search Tree*

- Developed ML-based classifier for online forum posts (Piazza), achieving 90% accuracy on labeled datasets

Out-of-Order RISC-V Microprocessor | *SystemVerilog, RISC-V, Python*

- Designed 32-bit out-of-order RISC-V processor with 2-way superscalar execution achieving 12.87ns critical path and 15% IPC improvement over normal 5-stage pipeline. Developed verification framework using **SystemVerilog** testbenches and Python automation
- Constructed modular and configurable GUI visual debugger based on Python (Tkinter) and VCD file parsing

NASA Mars Rover (Prototype) | *C++, Arduino Uno, Arduino IDE, 3D Printing*

- Led team of 4 in prototyping model of NASA Mars Rover within design constraints using rapid prototyping, engineering design standards, and requirements management. Implemented collision avoidance and functional robotic arm systems
- Designed, 3D printed, and tested multi-terrain wheels optimized for stability and traction across various surface conditions

Accelerometer Data Logger | *C++, Arduino IDE, ESP32, PCB*

- Designed and assembled **ESP32**-based PCB for 3-axis acceleration logging and real-time upload to Google Sheets for data acquisition

LEADERSHIP

National Gaming Competition

- Founded and led 120-member team in securing 2nd place in three-month global gaming competition among hundreds of teams

Student Leader

- Led community service event as Student Leader, Early College Alliance at Eastern Michigan University

SKILLS

Languages	C/C++, Python, TypeScript/JavaScript, Bash, SQL, MATLAB, SystemVerilog, Excel VBA
Tech	Next.js, React, LangGraph, OpenAI/Claude API, Docker, Linux (Arch/Ubuntu), ESP32, Raspberry Pi, CAN bus, 3D printing, object-oriented programming, flow/stress simulations, algorithm optimization, soldering
Tools	Git, Jira, Verdi, SolidWorks, Simulink, LaTeX, Arduino IDE, oscilloscope, unit testing frameworks, Tinkercad
Interests	AI, software/systems engineering, video game development, performance vehicles, psychology, rock climbing, nature